

CUMULATIVE DESIGN DOSSIER

Part I: Physical Foundation

- **LOOP-01:** Loop Charter & Boundaries
- **REQ-01:** τ_{world} , P_{99} Latency, d_{stop} Ledger
- **RUN-01:** Multi-Rate Skeleton & Watchdogs

Part II: 5 Sensory-Motor Organs

- **OBS-01:** DMA Ring Buffer Observation Contract
- **STATE-01:** $SE(3)$ Frame Tree & Timing Model
- **INTENT-01:** 3D Goal Grounding & Leases t_{expire}
- **PLAN-01:** Action Chunks (H) & C^2 Jerk
- **ENF-01:** 1 kHz MCU CBF Enforcer ($h(x) \geq 0$)

Part III: Integration & Release

- **PLACE-01:** Silicon Ledger (FLOPs, SRAM, W)
- **AUTH-01:** Bumpless Override & Lineage
- **REL-01:** ODD Specification & CAE Safety Case

4-EXPERT RED-TEAM AUDIT

Embedded Silicon & Real-Time Lead

- Zero-allocation MCU determinism (0 bytes dynamic)
- Watchdog timeout ≤ 50 ms; DMA memory contention

Embodied ML & Planning Lead

- 3D Spatial grounding; token latencies; model abstention
- Action chunking (H) with C^2 temporal continuity

Robotics & System Safety Lead

- Dynamic stopping bounds ($d_{\text{stop}} \leq d_{\text{clearance}}$)
- ISO 13850 stop categories; Claim-Argument-Evidence

Student & Practitioner UX Lead

- Zero-magic reproducibility; 12-min fault triage
- Hypothesis $\rightarrow$ Bisect $\rightarrow$ Confirm diagnostic logs

THE RELEASE VERDICT

DEPLOY

Unconstrained In-ODD Release

100% fault containment · Zero invariant violations · $P_{99,9} \leq \text{budget}$

CONDITION

Restricted Operating Envelope

Limited velocity ($v \leq 50\%$) · Reduced payload · Supervisor in line-of-sight

REFUSE

Shipment Blocked / Redesign

Hardware invariant breach · Uncontained fault · Tail latency exceeds stopping margin